

Graphical relationships and Inverse Functions - Solutions

Warm-up

Question 1:

To find the inverse function $f^{-1}(x)$, we replace $f(x)$ with y and solve for x in terms of y .

$$\begin{aligned}y &= 3x - 5 \\y + 5 &= 3x \\ \frac{y + 5}{3} &= x\end{aligned}$$

Now, we swap the variables x and y to express the inverse function in terms of x :

$$f^{-1}(x) = \frac{x + 5}{3}$$

Question 2:

The reciprocal function is defined as $y = \frac{1}{f(x)}$. Given $f(x) = \frac{1}{x+2}$, we substitute this expression into the reciprocal formula:

$$\begin{aligned}y &= \frac{1}{\frac{1}{x+2}} \\y &= x + 2\end{aligned}$$

Question 3:

For any one-to-one function $f(x)$, if a point (a, b) lies on the graph of $y = f(x)$, then the point (b, a) must lie on the graph of its inverse $y = f^{-1}(x)$.

Given the point $(3, -1)$ is on $y = f(x)$, the corresponding point on $y = f^{-1}(x)$ is:

$$(x, y) = (-1, 3)$$

Question 4:

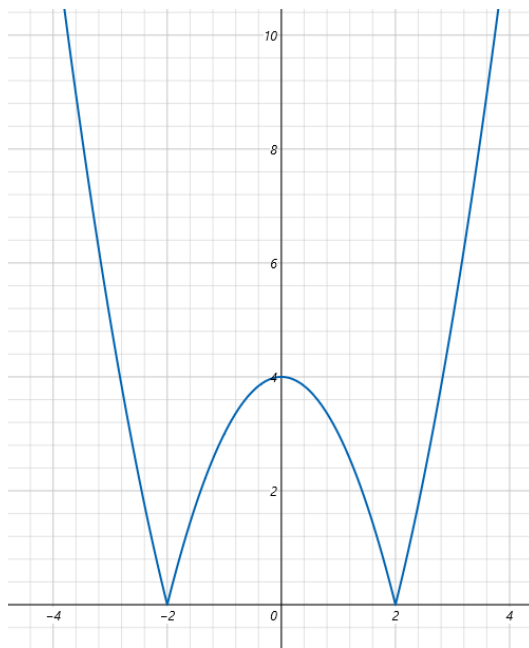
To sketch $y = |x^2 - 4|$, we start with the base graph $y = x^2 - 4$, which is a parabola opening upwards with x -intercepts at $x = \pm 2$ and a y -intercept at $(0, -4)$.

Applying the absolute value to the entire function reflects any part of the graph where $y < 0$ across the x -axis.

$$\text{For } x^2 - 4 \geq 0, \quad y = x^2 - 4$$

$$\text{For } x^2 - 4 < 0, \quad y = -(x^2 - 4) = 4 - x^2$$

Thus, the graph is a 'W' shape with vertices at $(\pm 2, 0)$ and a local maximum at $(0, 4)$.



Question 5:

To sketch $y = (|x - 2|)(|x + 1|)(|x|)$, we first consider the base function $g(x) = (x - 2)(x + 1)x$.
Expanding $g(x)$:

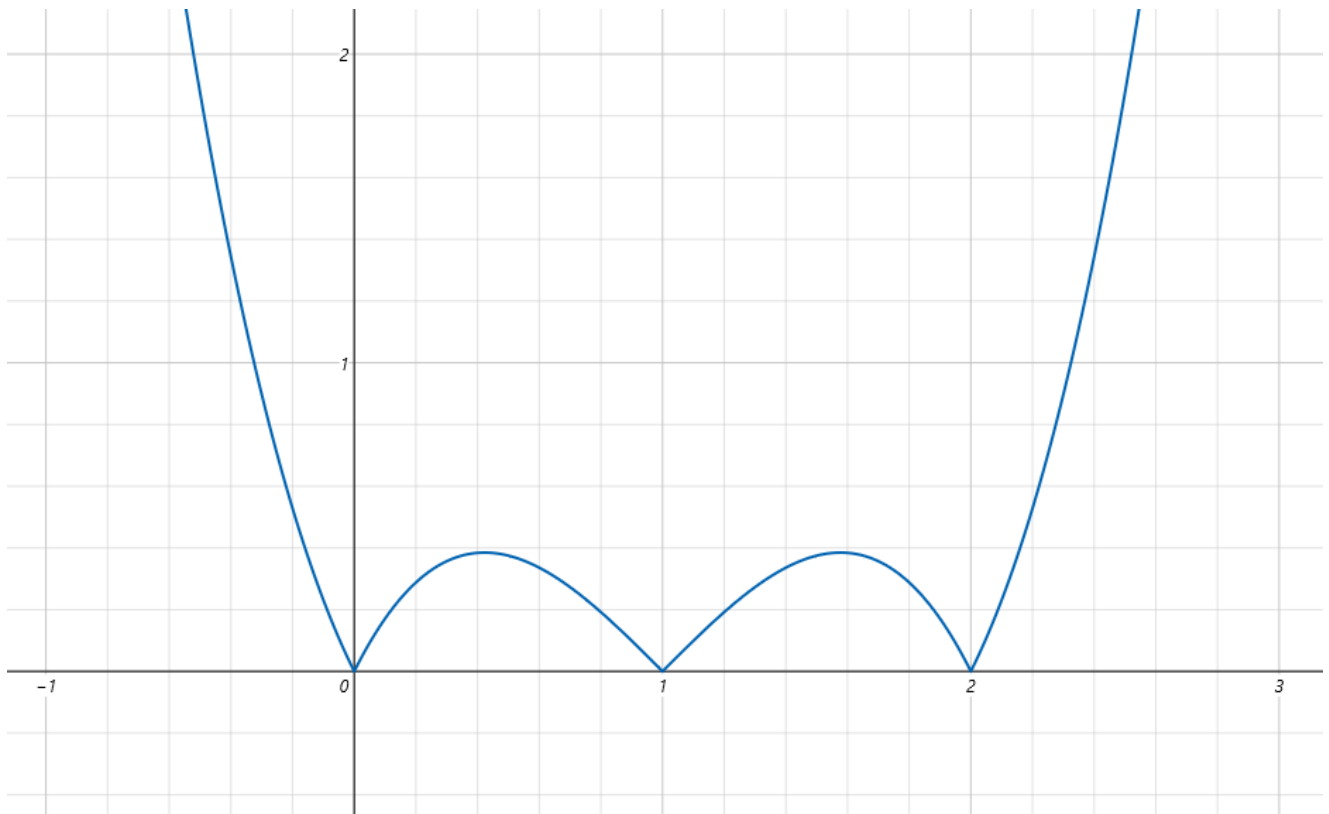
$$\begin{aligned} g(x) &= (x^2 + x - 2x - 2)x \\ &= (x^2 - x - 2)x \\ &= x^3 - x^2 - 2x \end{aligned}$$

The base graph $g(x)$ is a cubic with intercepts at $x = -1, 0, 2$.

Since $y = |g(x)|$ is not the case here, but rather the absolute value of each factor, we note that $|a||b||c| = |abc|$. Therefore, the function is equivalent to:

$$y = |(x - 2)(x + 1)x|$$

This means any part of the base cubic $g(x)$ that lies below the x -axis is reflected upwards. The resulting graph will have x -intercepts at $x = -1, 0, 2$ and all y -values will be non-negative.

**Question 6:**

The domain of a square root function $f(x) = \sqrt{g(x)}$ is defined for all x such that the radicand is non-negative:

$$\begin{aligned} x - 3 &\geq 0 \\ x &\geq 3 \end{aligned}$$

In interval notation, the domain is $[3, \infty)$.

Question 7:

To find the inverse $f^{-1}(x)$ for $f(x) = 4x + 7$, we set $y = 4x + 7$ and solve for x :

$$\begin{aligned} y &= 4x + 7 \\ y - 7 &= 4x \\ \frac{y - 7}{4} &= x \end{aligned}$$

Swapping x and y gives:

$$f^{-1}(x) = \frac{x - 7}{4}$$

Question 8:

The reciprocal function is $y = \frac{1}{f(x)}$. Given $f(x) = x - 3$:

$$y = \frac{1}{x - 3}$$

A vertical asymptote occurs where the denominator is zero and the numerator is non-zero:

$$\begin{aligned} x - 3 &= 0 \\ x &= 3 \end{aligned}$$

The vertical asymptote is $x = 3$.

Question 9:

For a one-to-one function $f(x)$, if the point (a, b) is on the graph of $y = f(x)$, then the point (b, a) is on the graph of $y = f^{-1}(x)$.

Given the point $(-2, 5)$ is on $f(x)$, the corresponding point on $f^{-1}(x)$ is:

$$(-2, 5) \implies (5, -2)$$

Question 10:

The domain of $f(x) = \sqrt{6 - 2x}$ requires the expression inside the square root to be non-negative:

$$\begin{aligned} 6 - 2x &\geq 0 \\ -2x &\geq -6 \\ x &\leq 3 \end{aligned}$$

Thus, the domain is $(-\infty, 3]$.

Question 11:

The composite function $f(f(x))$ is found by substituting $f(x)$ into itself. Given $f(x) = 2x - 3$:

$$\begin{aligned}
 f(f(x)) &= 2(f(x)) - 3 \\
 &= 2(2x - 3) - 3 \\
 &= 4x - 6 - 3 \\
 &= 4x - 9
 \end{aligned}$$

Question 12:

For any function f and its inverse f^{-1} , the domain of f is the range of f^{-1} , and the range of f is the domain of f^{-1} .

Given: - Domain of $f(x) = [-1, 4)$ - Range of $f(x) = (2, 7]$

$$\text{Domain of } f^{-1}(x) = \text{Range of } f(x) = (2, 7]$$

$$\text{Range of } f^{-1}(x) = \text{Domain of } f(x) = [-1, 4)$$

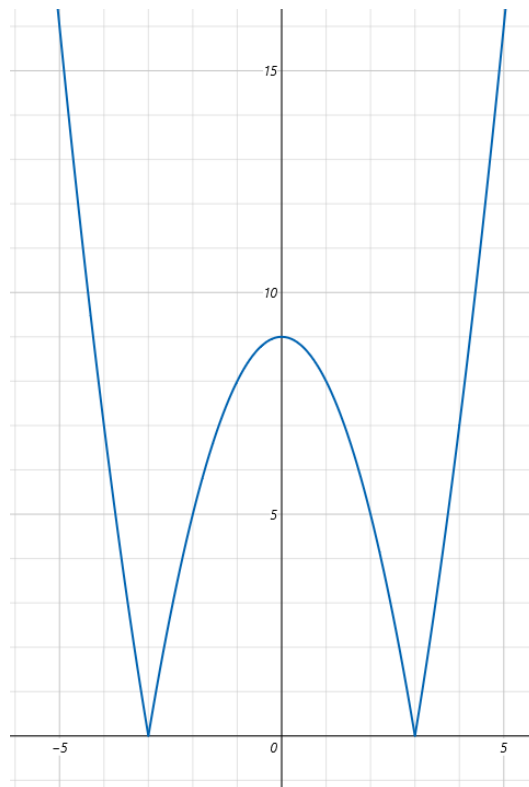
Question 13:

To sketch $y = |x^2 - 9|$, we first sketch $y = x^2 - 9$. This is a parabola with x -intercepts at $x = \pm 3$ and y -intercept at $(0, -9)$.

Applying the absolute value reflects the negative portion (between $x = -3$ and $x = 3$) above the x -axis.

Intercepts: $(\pm 3, 0)$ and $(0, 9)$

The graph is a 'W' shape with a local maximum at $(0, 9)$ and minimums at $(\pm 3, 0)$.



Question 14:

For the rational function $y = \frac{1}{x^2-4}$, asymptotes are found as follows:

Vertical asymptotes occur where the denominator is zero:

$$\begin{aligned}x^2 - 4 &= 0 \\(x - 2)(x + 2) &= 0 \\x &= \pm 2\end{aligned}$$

Horizontal asymptotes occur as $x \rightarrow \pm\infty$. Since the degree of the denominator is greater than the degree of the numerator:

$$\lim_{x \rightarrow \infty} \frac{1}{x^2 - 4} = 0$$

Thus, the horizontal asymptote is $y = 0$.

Question 15:

The function $y = |x - 3| - 1$ is a transformation of $y = |x|$.

1. Shift right by 3 units: $y = |x - 3|$. 2. Shift down by 1 unit: $y = |x - 3| - 1$.

Vertex: $(3, -1)$.

y -intercept (set $x = 0$):

$$y = |0 - 3| - 1$$

$$y = 3 - 1 = 2$$

x -intercepts (set $y = 0$):

$$0 = |x - 3| - 1$$

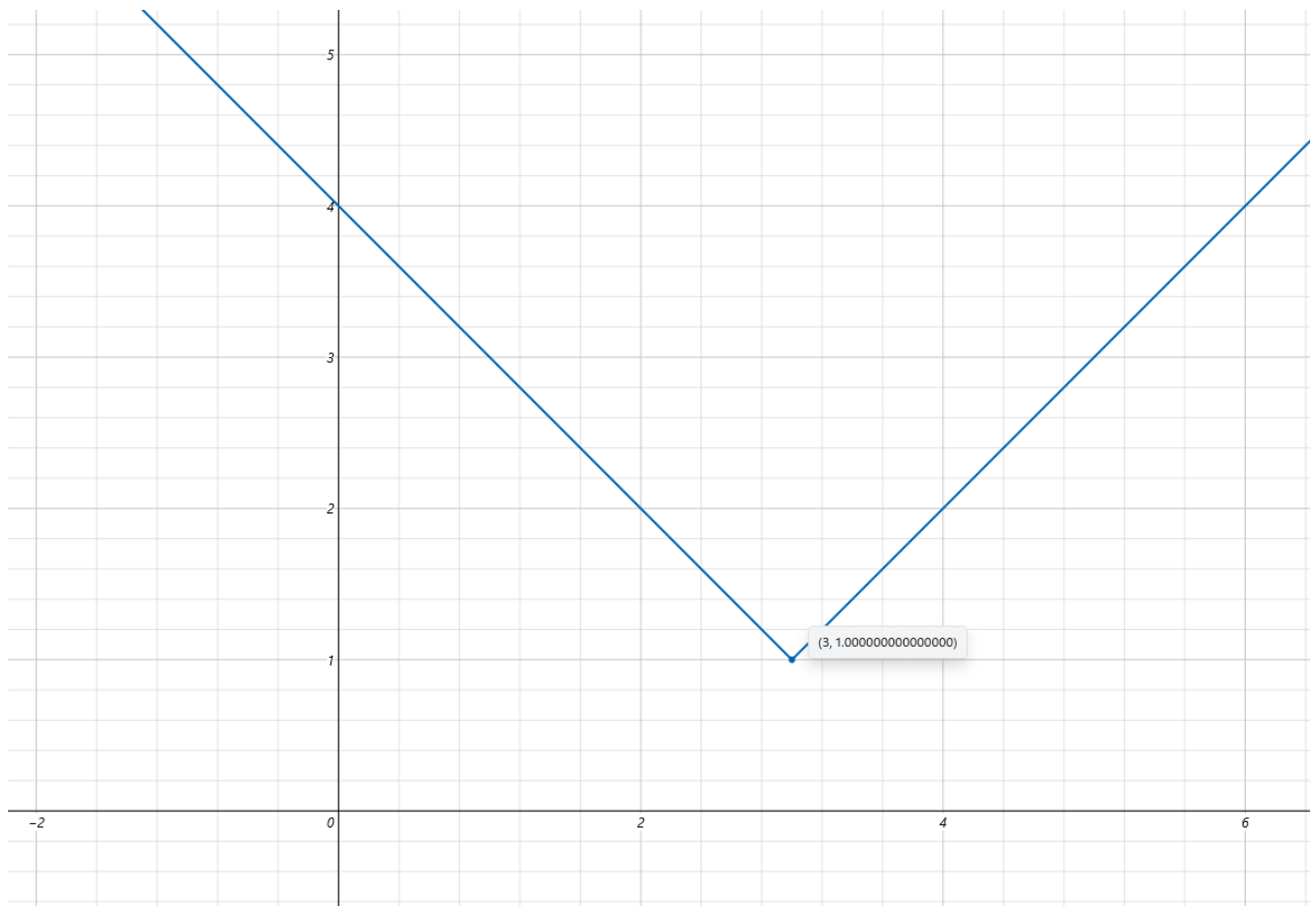
$$1 = |x - 3|$$

$$\pm 1 = x - 3$$

$$x = 3 \pm 1$$

$$x = 2, 4$$

Coordinates: Vertex $(3, -1)$, y -intercept $(0, 2)$, x -intercepts $(2, 0)$ and $(4, 0)$.

**Question 16:**

A function possesses a unique inverse if and only if it is one-to-one (passes the horizontal line test).

For $f(x) = x^2 + 4$, the graph is a parabola opening upwards with vertex at $(0, 4)$.

Domain restriction: $x \geq 0$

On the interval $[0, \infty)$, the function is strictly increasing. Therefore, it is one-to-one and possesses a unique inverse function.

Skill-building

Question 1:

We are given $f(x) = \frac{1-x}{1+x}$. To show $f(f(x)) = x$, we substitute $f(x)$ into itself:

$$\begin{aligned} f(f(x)) &= \frac{1 - f(x)}{1 + f(x)} \\ &= \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}} \\ &= \frac{\frac{(1+x)-(1-x)}{1+x}}{\frac{(1+x)+(1-x)}{1+x}} \\ &= \frac{1 + x - 1 + x}{1 + x + 1 - x} \\ &= \frac{2x}{2} \\ &= x \end{aligned}$$

Since $f(f(x)) = x$, the function is its own inverse. This implies that:

$$f(x) = f^{-1}(x)$$

Question 2:

Given $f(x) = (x - 2)(x - 4)$, we want to sketch $y = \frac{1}{f(|x|)}$.

First, let $g(x) = f(|x|) = (|x| - 2)(|x| - 4)$.

Vertical Asymptotes: $f(|x|) = 0$

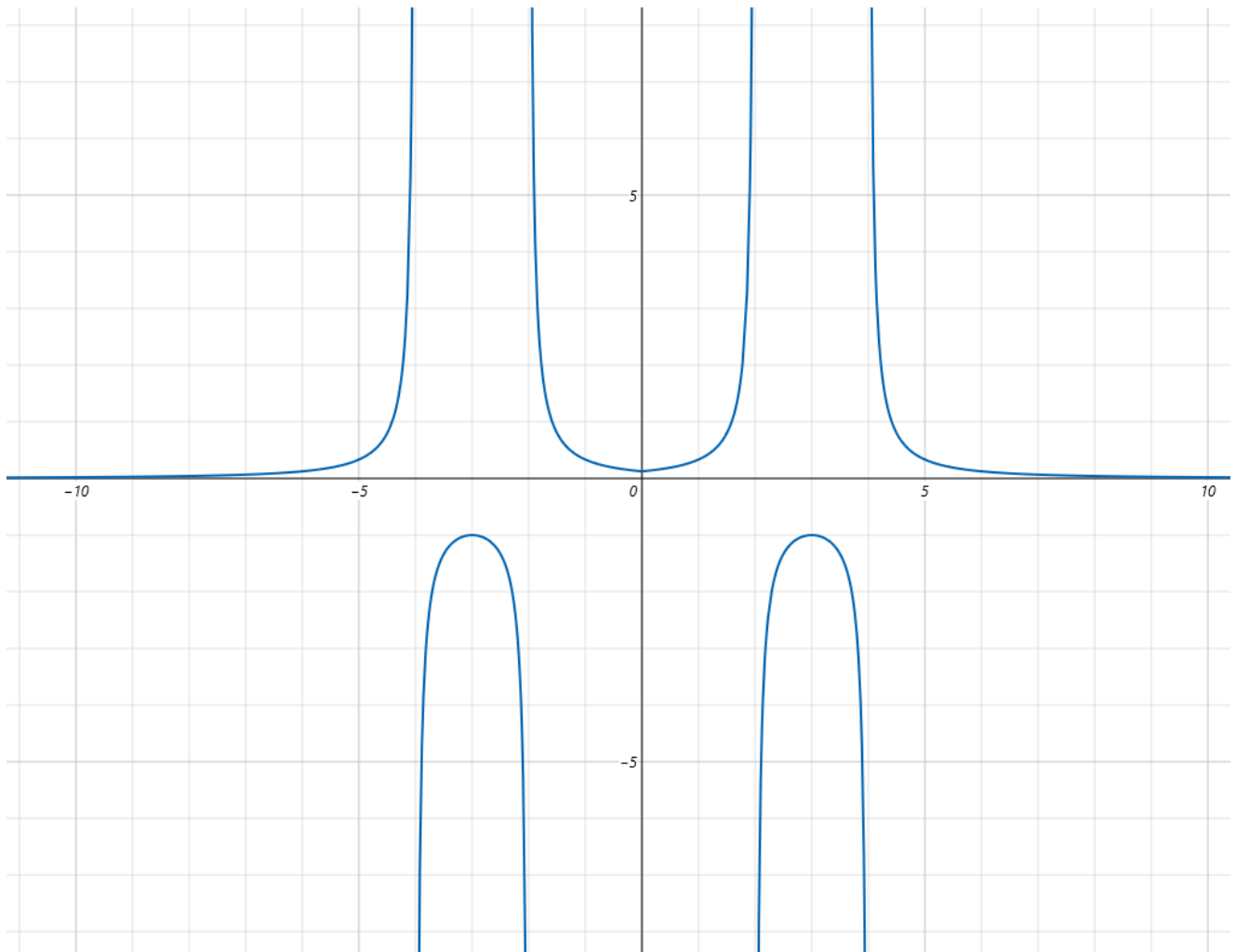
$$(|x| - 2)(|x| - 4) = 0$$

$$|x| = 2 \text{ or } |x| = 4$$

$$x = \pm 2, \pm 4$$

Horizontal Asymptote: $\lim_{x \rightarrow \pm\infty} \frac{1}{(|x| - 2)(|x| - 4)} = 0$

Therefore, $y = 0$

**Question 3a:**

Let $y = x + \sqrt{x^2 + 1}$. To find the inverse, we solve for x :

$$\begin{aligned}
y - x &= \sqrt{x^2 + 1} \\
(y - x)^2 &= x^2 + 1 \\
y^2 - 2xy + x^2 &= x^2 + 1 \\
-2xy &= 1 - y^2 \\
x &= \frac{y^2 - 1}{2y}
\end{aligned}$$

Swapping x and y to get the inverse function:

$$f^{-1}(x) = \frac{x^2 - 1}{2x}$$

Question 4:

For a function to have an inverse, it must be one-to-one (monotonic). We examine the derivative of $f(x) = x^4 - 4x^2 + 3$ to find the turning points:

$$\begin{aligned}
f'(x) &= 4x^3 - 8x \\
0 &= 4x(x^2 - 2) \\
x &= 0, \pm\sqrt{2}
\end{aligned}$$

The function is monotonic on the intervals $(-\infty, -\sqrt{2}]$, $[-\sqrt{2}, 0]$, $[0, \sqrt{2}]$, and $[\sqrt{2}, \infty)$.

We are restricted to the domain $x \leq k$. For the function to be one-to-one on this domain, k must not exceed the first turning point encountered from the left:

$$k \leq -\sqrt{2}$$

Thus, the maximum possible value of k is $-\sqrt{2}$.

Question 5a:

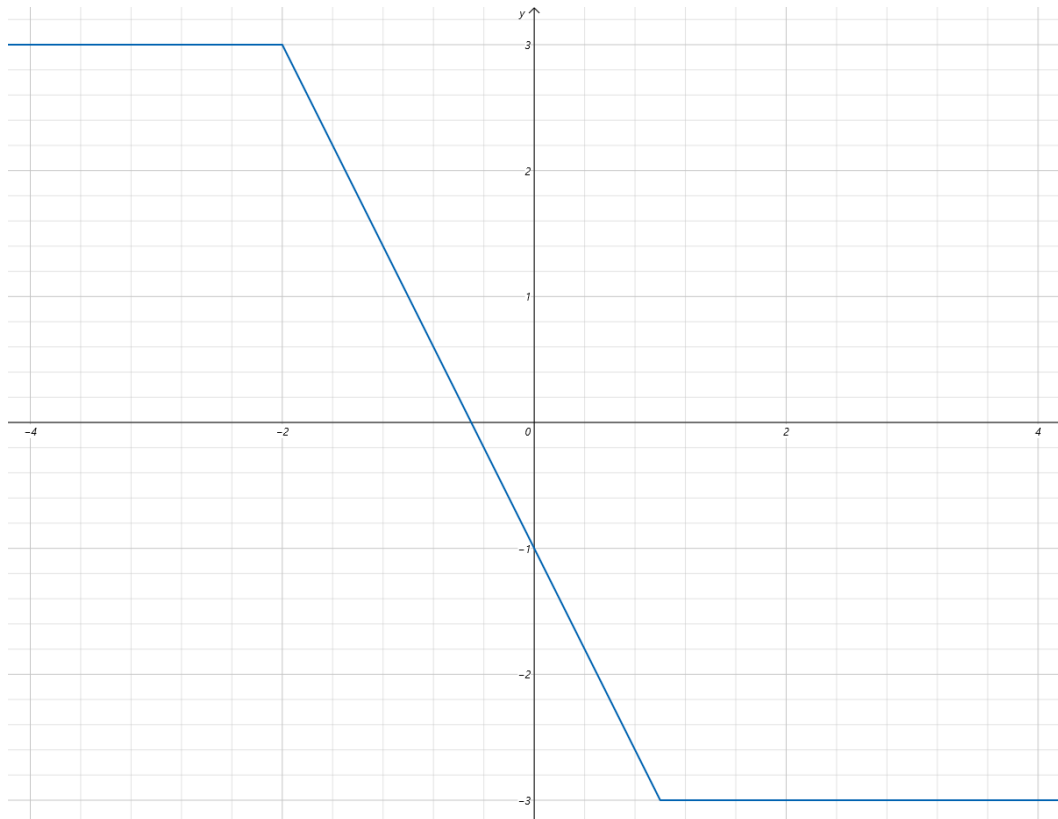
We analyze $f(x) = |x - 1| - |x + 2|$ by considering the critical points $x = 1$ and $x = -2$.

$$\text{Case 1: } x < -2 \implies f(x) = -(x - 1) - (-(x + 2)) = -x + 1 + x + 2 = 3$$

$$\text{Case 2: } -2 \leq x < 1 \implies f(x) = -(x - 1) - (x + 2) = -x + 1 - x - 2 = -2x - 1$$

$$\text{Case 3: } x \geq 1 \implies f(x) = (x - 1) - (x + 2) = x - 1 - x - 2 = -3$$

Graph: A horizontal line at $y = 3$ for $x < -2$, a linear segment from $(-2, 3)$ to $(1, -3)$, and a horizontal line at $y = -3$ for $x \geq 1$.



Question 5b:

To sketch $y = \frac{1}{f(x)}$, we use the results from 5a:

$$\text{For } x < -2, \quad y = \frac{1}{3}$$

$$\text{For } -2 < x < 1, \quad y = \frac{1}{-2x-1}$$

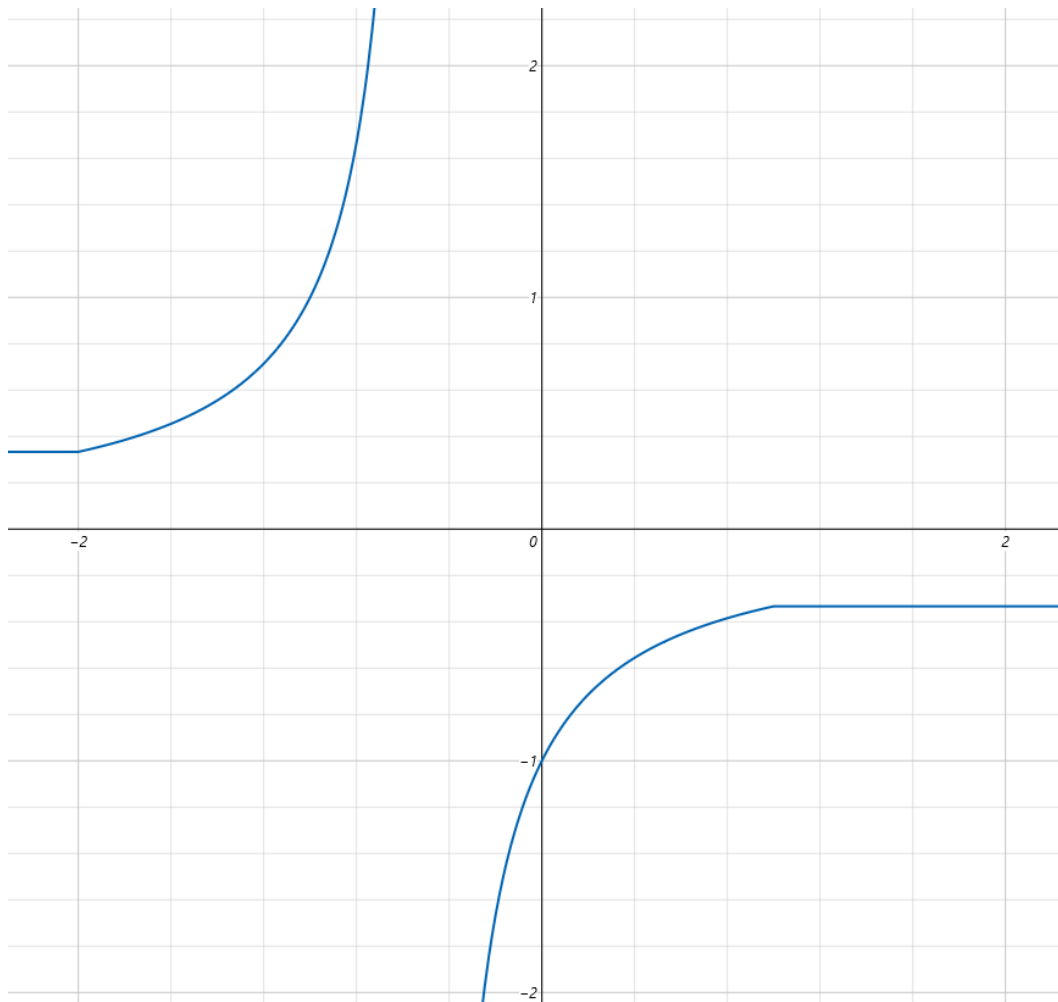
$$\text{For } x > 1, \quad y = \frac{1}{-3} = -\frac{1}{3}$$

Vertical asymptote occurs where $f(x) = 0$:

$$-2x - 1 = 0$$

$$x = -\frac{1}{2}$$

As $x \rightarrow -1/2^-$, $f(x) \rightarrow 0^+$, so $y \rightarrow \infty$. As $x \rightarrow -1/2^+$, $f(x) \rightarrow 0^-$, so $y \rightarrow -\infty$.

**Question 6:**

Let $y = \frac{x}{\sqrt{1-x^2}}$. To find the inverse, we solve for x :

$$\begin{aligned}
y^2 &= \frac{x^2}{1-x^2} \\
y^2(1-x^2) &= x^2 \\
y^2 - y^2x^2 &= x^2 \\
y^2 &= x^2(1+y^2) \\
x^2 &= \frac{y^2}{1+y^2} \\
x &= \frac{y}{\sqrt{1+y^2}} \text{ (since } x \text{ and } y \text{ have the same sign)}
\end{aligned}$$

Swapping x and y :

$$f^{-1}(x) = \frac{x}{\sqrt{1+x^2}}$$

Domain of $f^{-1}(x)$ is the range of $f(x)$. As $x \rightarrow 1^-$, $f(x) \rightarrow \infty$. As $x \rightarrow -1^+$, $f(x) \rightarrow -\infty$. Thus, the domain of $f^{-1}(x)$ is $\mathbb{R}$.

Range of $f^{-1}(x)$ is the domain of $f(x)$, which is $(-1, 1)$.

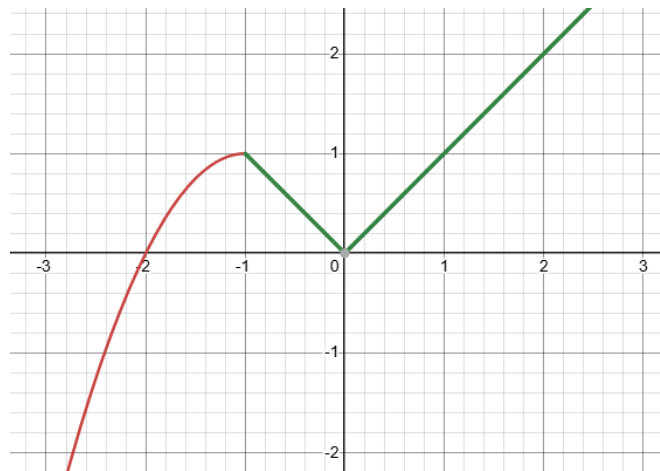
Question 7a:

We analyze the piecewise function $f(x)$:

$$\text{For } x < -1, \quad f(x) = -x^2 - 2x = -(x^2 + 2x) = -(x+1)^2 + 1$$

$$\text{For } x \geq -1, \quad f(x) = |x|$$

For $x < -1$, the graph is a parabola opening downwards with vertex at $(-1, 1)$. For $x \geq -1$, the graph is the absolute value function (V-shape) starting from $(-1, 1)$ and passing through $(0, 0)$ and $(1, 1)$.

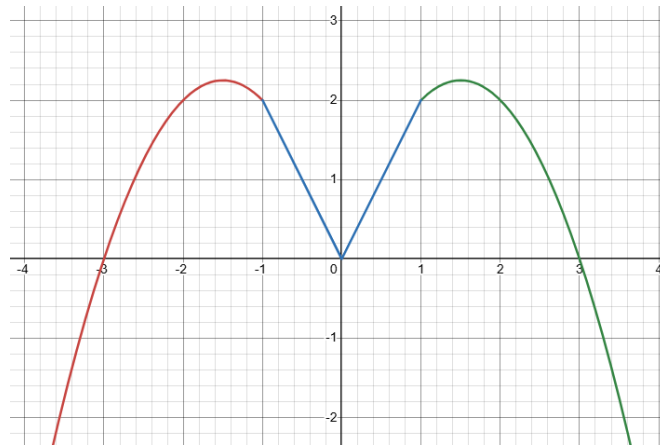


Question 7b:

Let $g(x) = f(x) + f(-x)$. We evaluate this for different intervals:

$$\begin{aligned}
&\text{If } x > 1, \quad f(x) = x, \quad f(-x) = -(-x)^2 - 2(-x) = -x^2 + 2x \\
&\implies g(x) = x - x^2 + 2x = -x^2 + 3x \\
&\text{If } -1 \leq x \leq 1, \quad f(x) = |x|, \quad f(-x) = |-x| = |x| \\
&\implies g(x) = 2|x| \\
&\text{If } x < -1, \quad f(x) = -x^2 - 2x, \quad f(-x) = |-x| = |x| = -x \\
&\implies g(x) = -x^2 - 2x - x = -x^2 - 3x
\end{aligned}$$

Intercepts: $g(0) = 0$. For $x > 1$, $-x^2 + 3x = 0 \implies x = 3$. For $x < -1$, $-x^2 - 3x = 0 \implies x = 0$ (not in range) or $x = -3$. Sharp edges at $x = \pm 1$.



Question 8:

For $f(x) = x^2 - 6x + 8$ to have a unique inverse on $x \leq k$, it must be one-to-one.

Find the vertex of the parabola:

$$x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$$

Since the parabola opens upwards, it is strictly decreasing on $(-\infty, 3]$. Therefore, the maximum value of k is 3.

Question 9:

Set $y = 2x + \sqrt{4x^2 + 1}$ and solve for x :

$$\begin{aligned}
y - 2x &= \sqrt{4x^2 + 1} \\
(y - 2x)^2 &= 4x^2 + 1 \\
y^2 - 4xy + 4x^2 &= 4x^2 + 1 \\
y^2 - 1 &= 4xy \\
x &= \frac{y^2 - 1}{4y}
\end{aligned}$$

Swapping x and y gives:

$$f^{-1}(x) = \frac{x^2 - 1}{4x}$$

Question 10:

Given $f(x) = x^2 - 4x + 3 = (x - 1)(x - 3)$.

Vertical asymptotes of $y = \frac{1}{f(x)}$ occur at $x = 1$ and $x = 3$.

y -intercept: set $x = 0$:

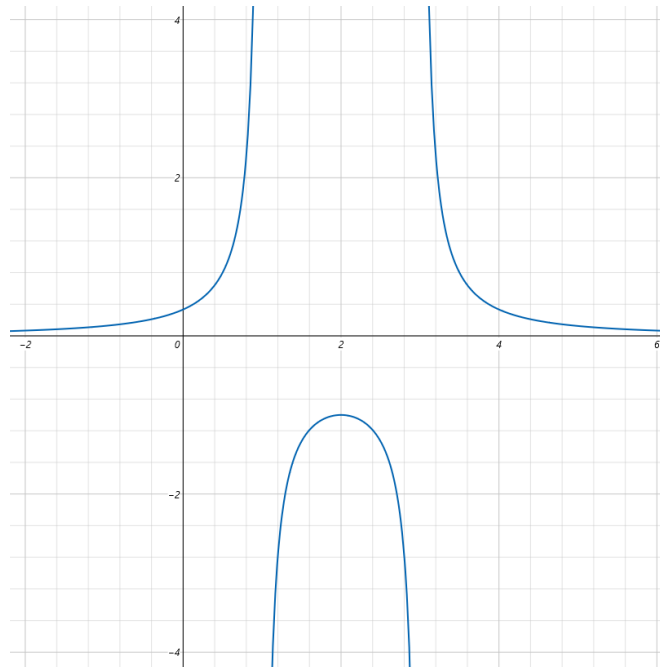
$$y = \frac{1}{0^2 - 4(0) + 3} = \frac{1}{3}$$

x -intercepts: None, as the numerator is a constant (1).

Turning point of $f(x)$ is at $x = -\frac{-4}{2} = 2$. At $x = 2$, $f(2) = 2^2 - 4(2) + 3 = -1$.

For the reciprocal function $y = \frac{1}{f(x)}$, the turning point is at $x = 2$ with $y = \frac{1}{-1} = -1$.

Since $f(x)$ has a minimum at $x = 2$, $1/f(x)$ has a local maximum at $(2, -1)$.



Question 11:

Set $y = \frac{x}{\sqrt{1+x^2}}$ and solve for x :

$$y^2 = \frac{x^2}{1+x^2}$$

$$y^2(1+x^2) = x^2$$

$$y^2 + y^2x^2 = x^2$$

$$y^2 = x^2(1 - y^2)$$

$$x^2 = \frac{y^2}{1 - y^2}$$

$$x = \frac{y}{\sqrt{1 - y^2}}$$

Swapping x and y gives:

$$f^{-1}(x) = \frac{x}{\sqrt{1-x^2}}$$

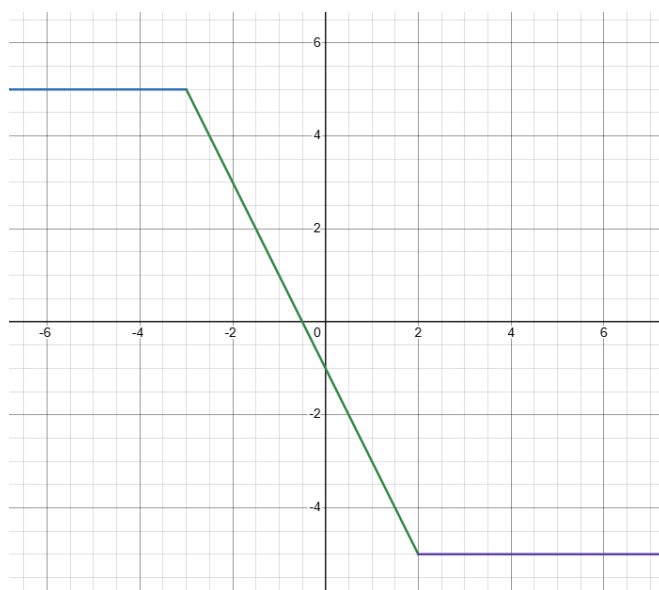
Domain of $f^{-1}(x)$ is the range of $f(x)$. Since $-1 < \frac{x}{\sqrt{1+x^2}} < 1$ for all x , the domain is $(-1, 1)$.
Range of $f^{-1}(x)$ is the domain of $f(x)$, which is $(-\infty, \infty)$.

Question 12:

For $f(x) = |x - 2| - |x + 3|$, we analyze the function in intervals:

1. $x < -3$: $f(x) = -(x - 2) - (-(x + 3)) = -x + 2 + x + 3 = 5$. 2. $-3 \leq x < 2$:
 $f(x) = -(x - 2) - (x + 3) = -x + 2 - x - 3 = -2x - 1$. 3. $x \geq 2$: $f(x) = (x - 2) - (x + 3) =$
 $x - 2 - x - 3 = -5$.

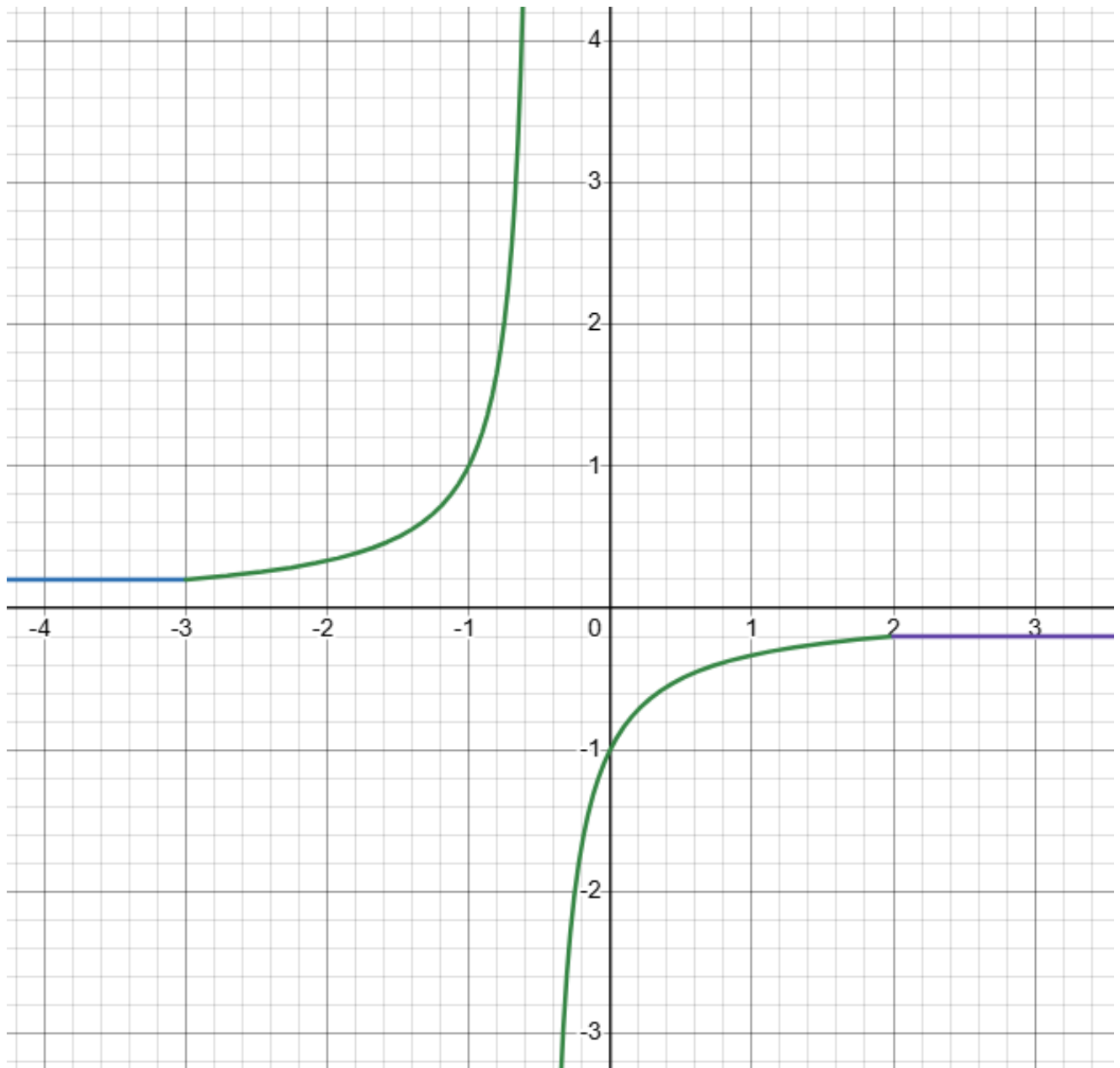
$$f(x) = \begin{cases} 5 & x < -3 \\ -2x - 1 & -3 \leq x < 2 \\ -5 & x \geq 2 \end{cases}$$



For $y = \frac{1}{f(x)}$, the reciprocal values are:

$$\frac{1}{f(x)} = \begin{cases} \frac{1}{5} & x < -3 \\ \frac{1}{-2x-1} & -3 \leq x < 2 \\ -\frac{1}{5} & x \geq 2 \end{cases}$$

Vertical asymptote for the reciprocal: $-2x - 1 = 0 \implies x = -0.5$.

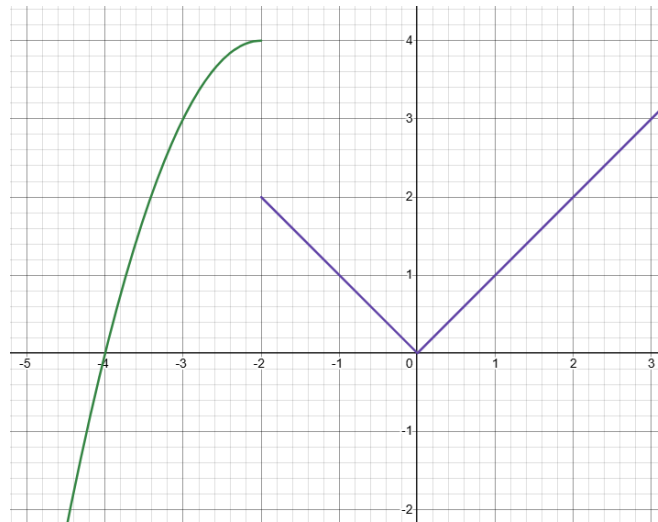


Question 13:

First, we sketch $f(x)$:

$$f(x) = \begin{cases} -x^2 - 4x & x < -2 \\ |x| & x \geq -2 \end{cases}$$

For $x < -2$, $f(x)$ is a parabola opening downwards with vertex at $x = -2, y = 4$. For $x \geq -2$, $f(x)$ is the absolute value function.



Now, we find $f(-x)$:

$$f(-x) = \begin{cases} -(-x)^2 - 4(-x) & -x < -2 \\ |-x| & -x \geq -2 \end{cases} \implies f(-x) = \begin{cases} -x^2 + 4x & x > 2 \\ |x| & x \leq 2 \end{cases}$$

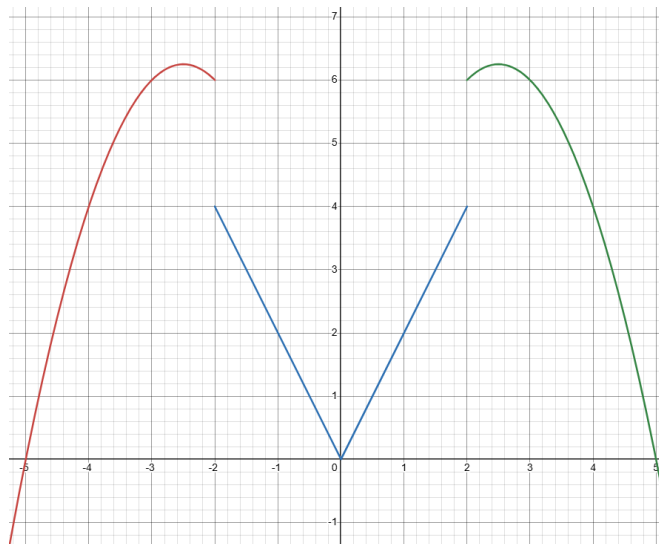
Now we sum $g(x) = f(x) + f(-x)$:

$$\text{For } x < -2: \quad f(x) = -x^2 - 4x, \quad f(-x) = -x \\ g(x) = -x^2 - 4x - x = -x^2 - 5x$$

$$\text{For } -2 \leq x < 2: \quad f(x) = |x|, \quad f(-x) = |x| \\ g(x) = |x| + |x| = 2|x|$$

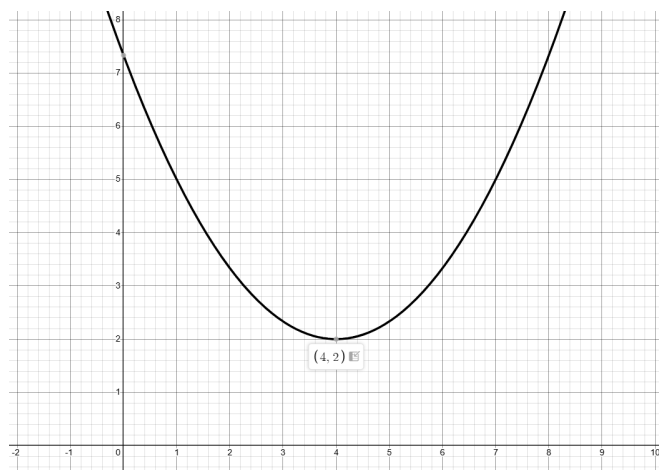
$$\text{For } x \geq 2: \quad f(x) = x, \quad f(-x) = -x^2 + 4x \\ g(x) = x - x^2 + 4x = -x^2 + 5x$$

Summary: $g(x) = -x^2 - 5x$ for $x < -2$, $2|x|$ for $-2 \leq x \leq 2$, and $-x^2 + 5x$ for $x > 2$.



Question 14:

Part (a): The function $f(x) = \frac{1}{3}(x - 4)^2 + 2$ is a parabola with vertex at $(4, 2)$.



Part (b): For the function to have an inverse, it must be one-to-one. The parabola is one-to-one on either side of the vertex $x = 4$. Since we need the domain to contain $x = 2026$, we must choose the right-hand branch:

Domain: $[4, \infty)$

Part (c): To find the inverse for $x \geq 4$, set $y = \frac{1}{3}(x - 4)^2 + 2$ and solve for x :

$$\begin{aligned}
 y - 2 &= \frac{1}{3}(x - 4)^2 \\
 3(y - 2) &= (x - 4)^2 \\
 \pm\sqrt{3(y - 2)} &= x - 4 \\
 x &= 4 \pm \sqrt{3(y - 2)}
 \end{aligned}$$

Since the domain is $x \geq 4$, we take the positive root:

$$x = 4 + \sqrt{3(y - 2)}$$

Swapping x and y gives:

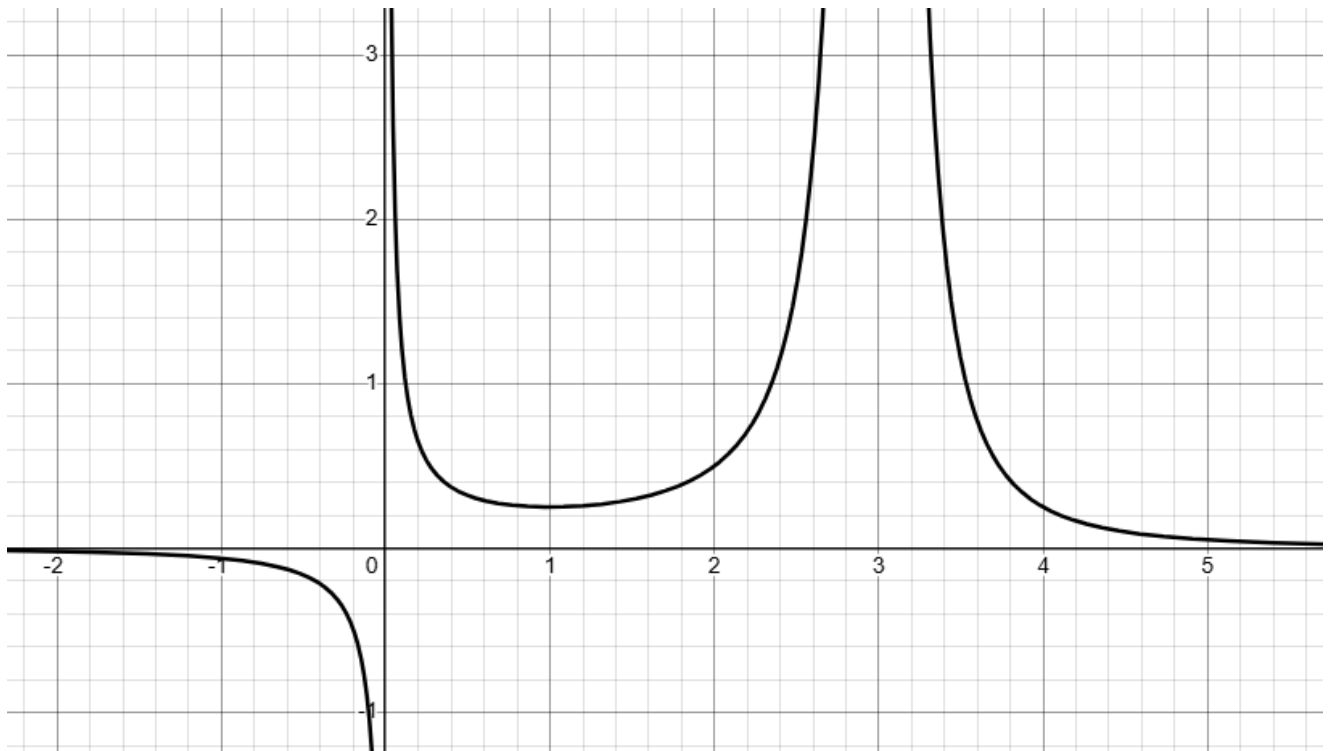
$$f^{-1}(x) = 4 + \sqrt{3(x - 2)}$$

Easier Exam Questions

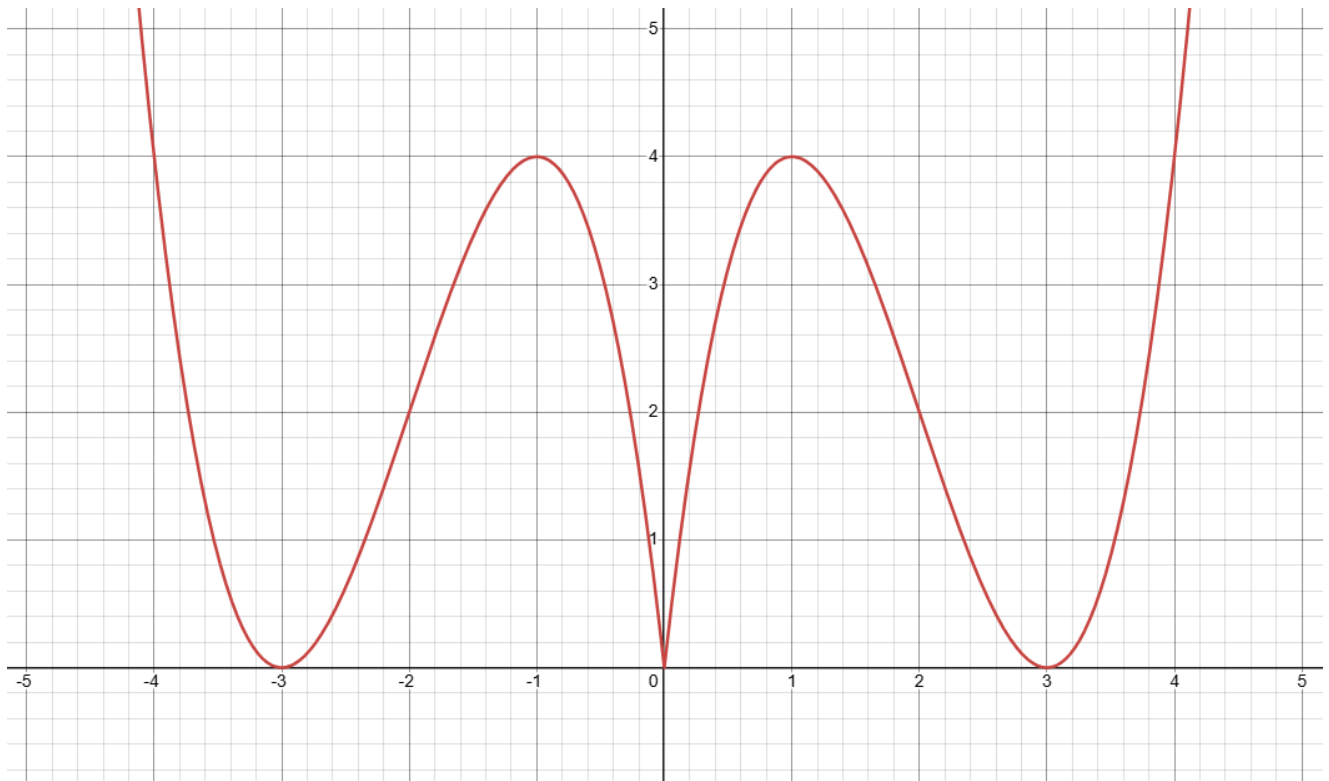
Baulkham Hills 2020 Q12b

Based on the provided graph of $f(x)$:

(i) For $y = \frac{1}{f(x)}$, the x -intercepts of $f(x)$ become vertical asymptotes. Where $f(x)$ is large, y is small. Where $f(x)$ is negative, y is negative. The graph will have vertical asymptotes at the roots of $f(x)$.



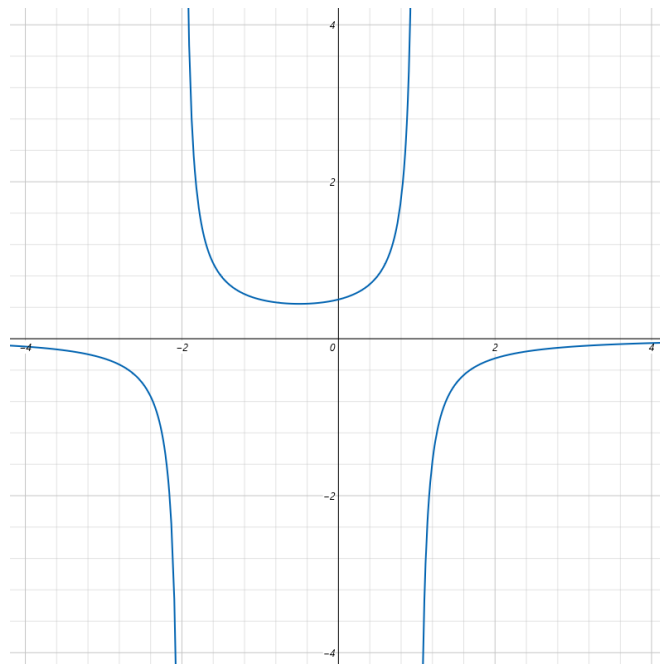
(ii) For $y = f(|x|)$, the part of the graph for $x < 0$ is discarded and the part for $x \geq 0$ is reflected across the y -axis to create a symmetric graph.



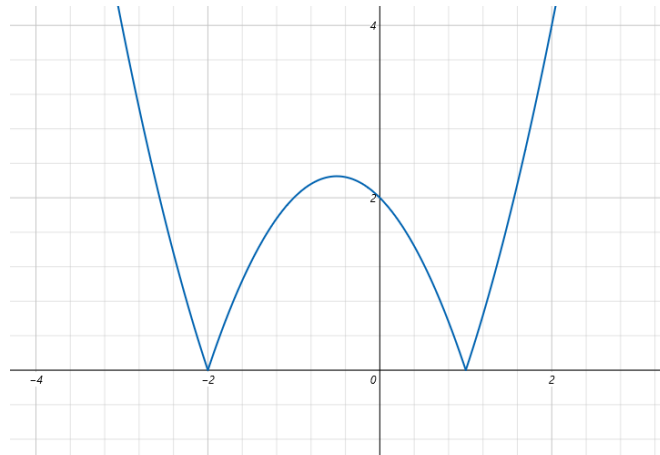
Girraween (2022) Q6c

Given $f(x) = (x + 2)(1 - x)$.

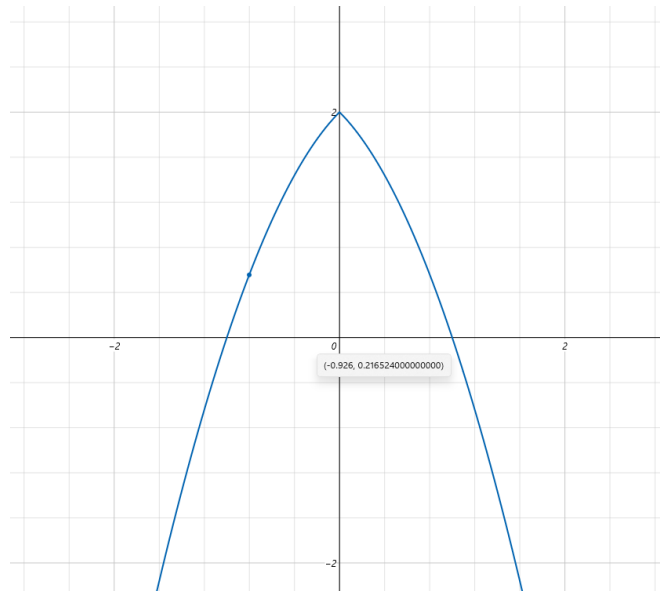
(i) $y = \frac{1}{f(x)} = \frac{1}{(x+2)(1-x)}$. Vertical asymptotes at $x = -2$ and $x = 1$. Horizontal asymptote at $y = 0$. Since $f(x)$ is a downward parabola, $1/f(x)$ will be positive between the roots and negative outside them.



(ii) $y = |f(x)| = |(x+2)(1-x)|$. The part of the parabola below the x -axis (where $x < -2$ or $x > 1$) is reflected upwards.



(iii) $y = f(|x|) = (|x|+2)(1-|x|)$. The part for $x < 0$ is replaced by the reflection of the part for $x \geq 0$. The x -intercepts are at $|x| = 1 \implies x = \pm 1$ and $|x| = -2$ (no solution).



James Ruse (2022) Q9

For the graph of $y = \frac{1}{ax^2+bx+a}$ to have exactly two asymptotes, one must be the horizontal asymptote $y = 0$ (since the degree of the denominator is higher than the numerator). The other two potential asymptotes are vertical asymptotes, which occur at the roots of the denominator $ax^2 + bx + a = 0$.

For exactly two asymptotes in total, there must be exactly one vertical asymptote. This happens if the quadratic $ax^2 + bx + a = 0$ has exactly one real root (a repeated root).

$$\begin{aligned}\Delta &= b^2 - 4(a)(a) = 0 \\ b^2 - 4a^2 &= 0 \\ (b - 2a)(b + 2a) &= 0 \\ b &= \pm 2a \\ \frac{a}{b} &= \pm \frac{1}{2}\end{aligned}$$

Looking at the options, $\frac{a}{b} = \frac{1}{2}$ is option (D).

Caringbah (2019) Q13b

Let $y = \frac{1-x}{3+x}$. To find the inverse, solve for x :

$$\begin{aligned} y(3+x) &= 1-x \\ 3y+xy &= 1-x \\ xy+x &= 1-3y \\ x(y+1) &= 1-3y \\ x &= \frac{1-3y}{y+1} \end{aligned}$$

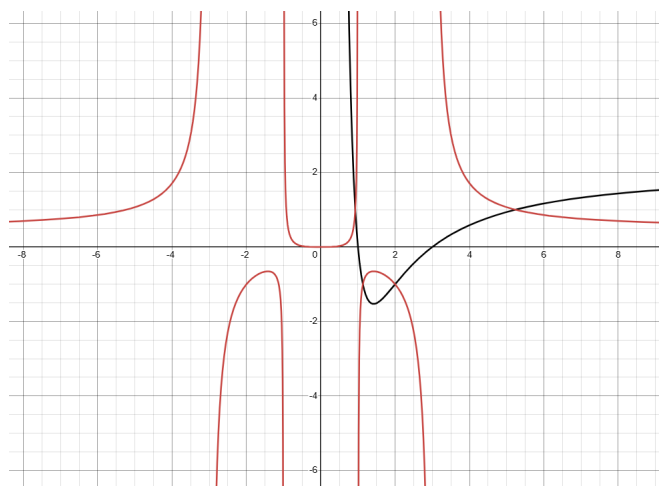
Swapping x and y :

$$f^{-1}(x) = \frac{1-3x}{x+1}$$

Harder Exam Questions

Baulkham Hills (2022) Q12a

To sketch $y = \frac{1}{f(|x|)}$, we first find $y = f(|x|)$ by reflecting the part of $f(x)$ for $x \geq 0$ across the y -axis. Then, we take the reciprocal of these values. Vertical asymptotes occur where $f(|x|) = 0$. Horizontal asymptotes occur as $x \rightarrow \pm\infty$. The image below shows $f(x)$ and the $\frac{1}{f(|x|)}$



Blacktown Boys (2023) Q13(e)

The function $f(x) = -|x|^2 + 4|x| - 3$ is an even function because $f(-x) = f(x)$.

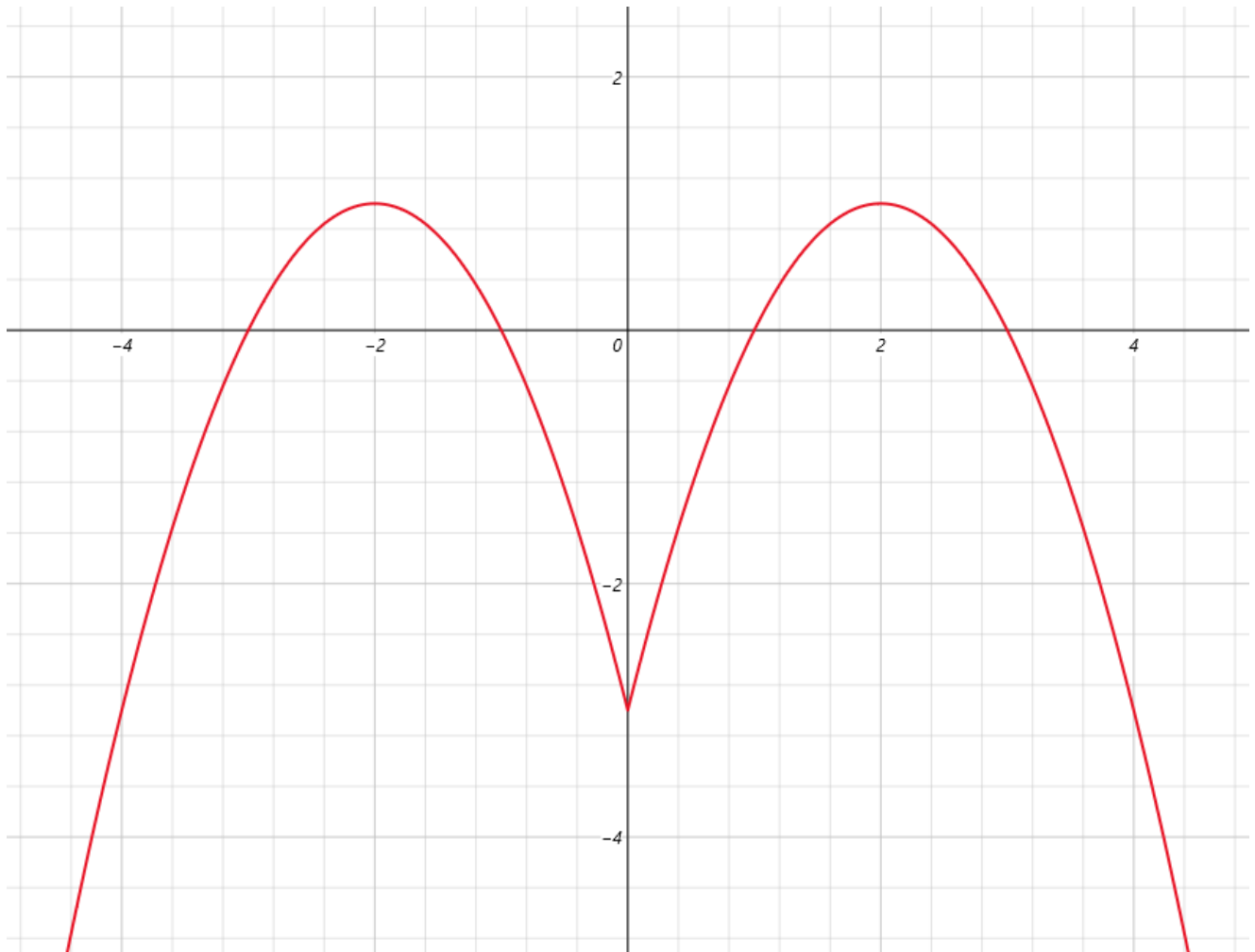
Let $g(x) = -x^2 + 4x - 3$ for $x \geq 0$.

$$g(x) = -(x^2 - 4x + 3) = -(x-1)(x-3)$$

For $x \geq 0$, the graph is a parabola opening downwards with roots at $x = 1$ and $x = 3$ and vertex at $x = 2, y = 1$.

Since $f(x)$ is even, we reflect this part across the y -axis to get the roots at $x = \pm 1, \pm 3$ and vertices at $x = \pm 2, y = 1$.

Intercepts: $(\pm 1, 0), (\pm 3, 0), (0, -3)$



Presbyterian (2022) Q12d

Given $f(x) = x + 2$ for $x < 0$ and $f(x) = |x - 2|$ for $x \geq 0$.

Let $g(x) = f(x) + f(-x) + 1$.

$$\begin{aligned} \text{If } x < 0, \quad f(x) &= x + 2, \quad f(-x) = |-x - 2| = |x + 2| \\ \implies g(x) &= (x + 2) + |x + 2| + 1 \end{aligned}$$

$$\begin{aligned} \text{If } x \geq 0, \quad f(x) &= |x - 2|, \quad f(-x) = -x + 2 \\ \implies g(x) &= |x - 2| + (-x + 2) + 1 = |x - 2| - x + 3 \end{aligned}$$

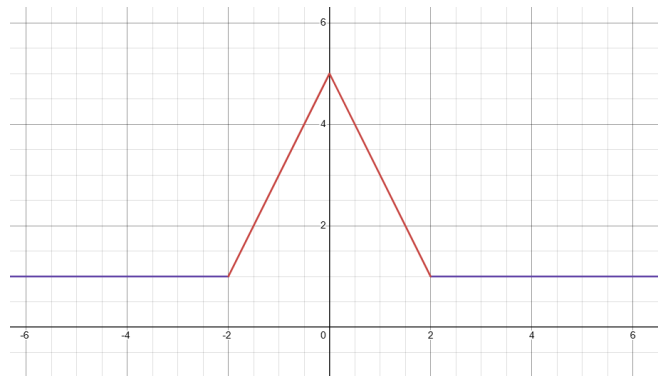
Analyzing $g(x)$ for $x < 0$:

$$\begin{aligned} \text{If } x < -2, \quad g(x) &= (x + 2) - (x + 2) + 1 = 1 \\ \text{If } -2 \leq x < 0, \quad g(x) &= (x + 2) + (x + 2) + 1 = 2x + 5 \end{aligned}$$

Analyzing $g(x)$ for $x \geq 0$:

$$\begin{aligned} \text{If } 0 \leq x < 2, \quad g(x) &= -(x - 2) - x + 3 = -x + 2 - x + 3 = -2x + 5 \\ \text{If } x \geq 2, \quad g(x) &= (x - 2) - x + 3 = 1 \end{aligned}$$

Intercepts: $g(0) = 5$. x -intercepts: $2x + 5 = 0 \implies x = -2.5$ (not in range). $-2x + 5 = 0 \implies x = 2.5$ (not in range). No x -intercepts. The graph is a trapezoidal shape with a flat top at $y = 1$ for $x \leq -2$ and $x \geq 2$, and a peak at $(0, 5)$.

**Fort St (2022) Q4**

For a one-to-one function f , the domain of f^{-1} is the range of f , and the range of f^{-1} is the domain of f .

Given:

$$\begin{aligned} \text{Domain of } f &= [0, 5) \\ \text{Range of } f &= (-2, 5] \end{aligned}$$

Therefore:

$$\begin{aligned} \text{Domain of } f^{-1} &= (-2, 5] \\ \text{Range of } f^{-1} &= [0, 5) \end{aligned}$$

This matches option (A).

Presbyterian (2023) Q1

For a one-to-one function f , the domain of f is the range of f^{-1} , and the range of f is the domain of f^{-1} .

Given: - Domain of $f(x) = [0, 5]$ - Range of $f(x) = (-2, 5]$

$$\text{Domain of } f^{-1}(x) = \text{Range of } f(x) = (-2, 5]$$

$$\text{Range of } f^{-1}(x) = \text{Domain of } f(x) = [0, 5]$$

This matches option (A).

Sydney Boys (2019) Q15d

Given $f(x) = \frac{1-x}{1+x}$. We find $f^2(x) = f(f(x))$:

$$\begin{aligned} f^2(x) &= \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}} \\ &= \frac{\frac{1+x-(1-x)}{1+x}}{\frac{1+x+(1-x)}{1+x}} \\ &= \frac{2x}{2} = x \end{aligned}$$

Since $f^2(x) = x$, the function is its own inverse. This means that applying the function twice returns the original input.

$$f^{2020}(x) = (f^2(x))^{1010} = x^{1010} \text{ (incorrect notation, this is an iteration)}$$

$$\text{Correct: } f^{2020}(x) = f^2(f^2(\dots f^2(x)\dots)) = x$$

Since 2020 is even, the sequence of iterations $f, f^2, f^3, f^4 \dots$ repeats every 2 steps. Since $f^2(x) = x$, then $f^{2n}(x) = x$ for any integer n . Thus, $f^{2020}(x) = x$.

Sydney Girls (2022) Q4a(i)

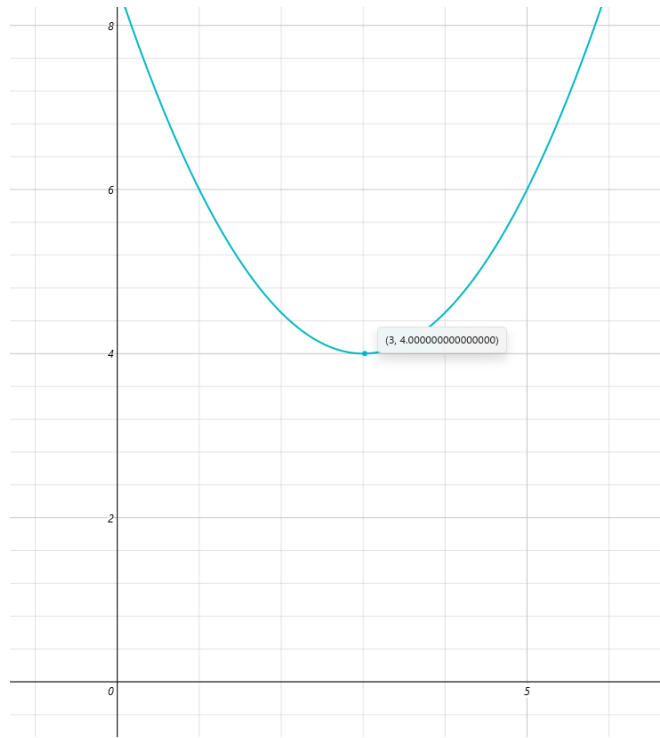
Given $f(x) = \frac{1}{2}(x-3)^2 + 4$.

(i) The vertex is at $(3, 4)$. The parabola opens upwards.

The largest domain containing $x = 2022$ for which the function is one-to-one is the right-hand branch of the parabola:

$$\text{Domain: } [3, \infty)$$

(ii) To find $f^{-1}(x)$ for $x \geq 3$:



$$\begin{aligned}
 y &= \frac{1}{2}(x-3)^2 + 4 \\
 y - 4 &= \frac{1}{2}(x-3)^2 \\
 2(y-4) &= (x-3)^2 \\
 \sqrt{2(y-4)} &= x-3 \\
 x &= 3 + \sqrt{2(y-4)}
 \end{aligned}$$

Swapping x and y :

$$f^{-1}(x) = 3 + \sqrt{2(x-4)}$$

(iii) Let $a < 3$. By symmetry of the parabola, $f(a) = f(3 + (3 - a)) = f(6 - a)$. Since $6 - a > 3$, the point $(6 - a, f(a))$ is in the domain of the inverse function. Thus:

$$f^{-1}(f(a)) = 6 - a$$

James Ruse (2024) Q5

The points where $f(x)$ and $f^{-1}(x)$ intersect are the points where $f(x) = x$ (if the function is increasing) or where $f(x) = -x$ (if the function is decreasing).

Given $f(x) = -\sqrt{1 - \sqrt{1 + x}}$.

$$\text{For } f(x) = x : \quad -\sqrt{1 - \sqrt{1+x}} = x$$

Since the LHS is non-positive, $x \leq 0$.

$$\text{Squaring both sides: } 1 - \sqrt{1+x} = x^2$$

$$\sqrt{1+x} = 1 - x^2$$

$$\text{Squaring again: } 1 + x = (1 - x^2)^2 = 1 - 2x^2 + x^4$$

$$\text{Simplifying: } x^4 - 2x^2 - x = 0$$

$$\text{Factoring: } x(x^3 - 2x - 1) = 0$$

$$\text{Testing roots: } x = -1 \text{ is a root of } x^3 - 2x - 1$$

$$\implies x(x+1)(x^2 - x - 1) = 0$$

Possible x values: $0, -1, \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}$.

Checking validity for $f(x) = x$:

- If $x = 0$, $f(0) = -\sqrt{1-1} = 0$. (Valid)
- If $x = -1$, $f(-1) = -\sqrt{1-0} = -1$. (Valid)
- If $x = \frac{1-\sqrt{5}}{2} \approx -0.618$, $f(-0.618) = -\sqrt{1 - \sqrt{0.382}} = -\sqrt{1 - 0.618} = -\sqrt{0.382} \approx -0.618$. (Valid)
- If $x = \frac{1+\sqrt{5}}{2} > 0$, invalid since $x \leq 0$.

There are 3 points of intersection on the line $y = x$. Thus, the answer is (C).

